IVÁN RICARDO NAVARRO ESPINOSA

Chemical Engineer, MSc | Process Optimization | Junior Data Scientist | Lean Six Sigma Green Belt
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PROFESSIONAL SUMMARY

Chemical Engineer, MSc, with 13 years of manufacturing experience in the chemical and food industries, coordinating production, improvement projects and teams of 20 people. Focused on process optimization through data analysis, predictive modeling and Lean Six Sigma. Using operational data, reduced vinegar fermentation time by 25%, increasing the section's production capacity by 15%. Data science training with projects in Python, SQL and Machine Learning, such as gold recovery prediction in a processing plant and customer churn prediction. Seeking roles in industrial analytics or data science applied to processes.

TECHNICAL SKILLS

Engineering & processes: Lean Six Sigma · PFD/P&ID process diagrams · Mass and energy balances · Root cause analysis · Pilot and industrial trials · Production KPIs · Management systems (ISO 9001, ISO 14001, ISO 45001, BRCGS, GMP).

Data science: Python (pandas, NumPy, scikit-learn, LightGBM, SciPy, statsmodels, Seaborn) · Machine Learning (regression and classification models) · Time series · SQL (PostgreSQL) · MATLAB · Jupyter Notebooks · Visual Studio Code · Git/GitHub.

Statistics & experimentation: Hypothesis testing · A/B testing · Design of experiments · Control charts · Multiple regression.

Visualization & industrial platforms: Excel · Power BI · SAP (production modules) · AVEVA PI System.

DATA SCIENCE PROJECTS

Full portfolio: *ivanrnavarro.notion.site*

Gold Recovery Prediction in a Purification Process — Zyfra | TripleTen, 2026 | View on GitHub

Analyzed 22,716 process records (87 variables) from a gold extraction and purification plant, validated the recovery calculation and cleaned anomalous sensor records. Trained one model per stage (linear regression for rougher and Random Forest for purification) with cross-validation and a custom weighted sMAPE metric. Achieved a final sMAPE of 9.73% on the test set and identified silver losses in tailings as a recovery opportunity. *Python · pandas · scikit-learn · Seaborn*

Customer Churn Prediction — Interconnect | TripleTen Capstone Project, 2026 | View on GitHub

Integrated 4 data sources covering 7,043 customers and detected data leakage in billing variables before modeling. Compared logistic regression, Random Forest and LightGBM with class-imbalance handling, hyperparameter tuning and a decision threshold optimized for Recall. LightGBM reached an AUC-ROC of 0.91 and detected 89% of churning customers; contacting the 30% highest-risk customers captures 76% of churn. *Python · pandas · scikit-learn · LightGBM · Seaborn*

Hourly Demand Forecasting — Sweet Lift Taxi | TripleTen, 2026 | View on GitHub

Resampled 6 months of taxi orders recorded every 10 minutes into hourly intervals, decomposed the series and built calendar features, 48 lags and a rolling mean. Compared linear regression, decision tree and Random Forest using time-series cross-validation (TimeSeriesSplit). Random Forest achieved an RMSE of 41.4 orders per hour, 30% better than a naive baseline (59) and below the target of 48. *Python · pandas · statsmodels · scikit-learn*

PROFESSIONAL EXPERIENCE

Data Science Training — TripleTen | Remote | January – November 2026

Intensive bootcamp in Python, SQL, statistics and Machine Learning applied to processes and business.

Sucroal S.A. (formerly Sucromiles S.A.) | Palmira, Valle del Cauca | April 2012 – December 2025

Chemical and food manufacturing plant (plasticizers, natural vinegar/acetic acid, alcohol, solvents) with continuous 24/7 operation, serving domestic and export markets.

Production Leader — *January 2021 – December 2025*

Plasticizer and natural vinegar/acetic acid lines, leading teams of 8 to 10 operators.

- Reduced vinegar fermentation time by 25%, increasing section capacity by 15%, through operational data analysis, root cause analysis and PDCA cycles.

- Routinely analyzed plant historian (PI) data and time series of operating variables in Excel, tracking cost per kilogram and the yield of raw materials and utilities.
- Planned and scheduled short- and medium-term production, coordinating supply with purchasing and supply chain, as well as maintenance.
- Standardized operations with SOPs, digital forms and an Excel KPI dashboard (plan compliance, consumption, quality and productivity), reducing operational waste by 5%.
- Led the industrial validation of a new vinegar filtration plant (COP 800 million investment, ≈ USD 222,000), from pilot plant to production runs, and documented with yield data its failure to meet specifications, providing technical support for the claim against the supplier.
- Contributed to maintaining zero workplace accidents for more than 3 years, coordinating initiatives with maintenance and industrial safety.
- Since 2024, served as accompanying auditor in internal audits (2 to 3 per year) and supplier audits (1 to 2 per year); hosted customer and certification body audits with no critical nonconformities.

Production Coordinator — *January 2014 – December 2020*
Plasticizer, natural vinegar/acetic acid and alcohol lines, leading teams of 20 operators.

- Implemented improvements in acetyl tributyl citrate (ATBC) production using mass balances and time-and-motion studies, reducing production times by 10%.
- Contributed to the FCC certification of triethyl citrate (TEC) and to the certification of ATBC for export markets, increasing sales of both products by 20%.
- Coordinated production planning and scheduling to meet monthly plans and export programs, addressing process bottlenecks.
- Participated in implementing HACCP and food safety plans for the TEC FCC and vinegar lines, hosting BRCGS audits through the certification of these products.

Process Engineer — *April 2012 – December 2013*
Plasticizer line, leading a team of 4 operators.

- Technically validated the expansion of the plasticizer line (COP 1,500 million investment, ≈ USD 417,000), defining critical process parameters and monitoring them with control charts, which increased line capacity from 800 to 1,500 tons per year.
- Contributed to upgrading ATBC from industrial grade to specialty grade by adjusting operating conditions.

Adjunct Lecturer — Universidad del Valle and Unicomfacauca | *2009 – 2012*

EDUCATION

- **Data Science Bootcamp** — TripleTen | 2026
- **Master's in Chemical Engineering** — Universidad del Valle | 2012
- **B.S. in Chemical Engineering** — Universidad Nacional de Colombia, Manizales | 2005

CERTIFICATIONS & COURSES

- **Skills for Work (Excel)** — Coursera, Fundación Santander | 2026
- **Power BI** — Edutin | 2026
- **Internal Auditor ISO 9001, ISO 14001, ISO 45001, BRCGS** — Cencode | 2025
- **Lean Six Sigma Green Belt** — DSS+ | 2024

LANGUAGES

- **Spanish** — Native | **English** — Upper-intermediate (B2)